# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• (Touch)screen (1)<br>• Speaker (1)<br>• Vibration (motor) (1)<br>• Flashlight/LED light/torch (1)<br>• NFC chip (1) | **Accept:**<br>• display for screen<br>**Do not accept:**<br>• monitor<br>• printer<br>• headset<br>• other peripheral devices | **2** |
| Question | Answer | Additional Guidance | Mark |
| 2(b)(i) | **The only correct answer is D**<br><br>*A is not correct because cache is located closer to the CPU*<br>*B is not correct because random-access memory is a physical card on the motherboard*<br>*C is not correct because read-only memory cannot be written to* | | **1** |
| Question | Answer | Additional Guidance | Mark |
| 2(b)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• Often held/stored remotely/stored on the internet (1)<br>• Held/managed by a third party (1)<br>• Requires internet connectivity to access (1) | | **2** |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

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## Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award **one** mark for any of the following up to a maximum of **three** marks:<br>• Clock speed/number of cycles per second (1)<br>• Number of processor cores (1)<br>• Size of cache (memory) (1)<br>• Type of cache (1)<br>• Width of data bus/word length (1)<br>• Width of address bus (1)<br>• Whether the CPU is overclocked or not (1)<br>• Pipelining architecture (1)<br>• Thermal throttling / cooling system (reducing clock speed) (1) | **Accept:**<br>• units of clock speed e.g. GHz for MP1<br>• width of bus for either MP5 or MP6 but not both<br>**Do not accept:** cache without clarification of size or type | 3 |
| Question | Answer | Additional Guidance | Mark |
| 2(c)(ii) | Award up to **two** marks for a linked description such as:<br>• The CPU places an address on to the address bus (1) to specify the location in main memory (1)<br>• (The address bus) contains the location in RAM (1) to read from/write to (1)<br>• The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1)<br>• The address bus carries the memory location (1) of the current/next instruction (1) | Explanation can be a combination of different marking points. E.g. MP1a and MP2b | 2 |
| Question | Answer | Additional Guidance | Mark |
| 2(d) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• Sub-atomic particles (1)<br>• Qubit(s) (1)<br>• Entanglement (1) behaviour of a pair of qubits can influence each other<br>• Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously | **Do not accept:**<br>'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question | 2 |
| | | **Total for question 2** | **12** |

# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 1 of 5 · Question 2 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Cloud storage: what it is, pros, cons and security

Answer 2 of 5 · Question 2 part a · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award **one** from each:<br><br>Zafer:<br>1. Choose strong (1) authentication credentials (username/password)<br>2. Not sharing his credentials (1)<br>3. Not having automatic login to cloud / leaving his machine unattended (1)<br>4. Changing password regularly (1)<br>5. Password protect documents (1)<br><br>Provider:<br>1. Infrastructure (e.g. firewall/servers) must be secure from unauthorised access (1)<br>2. Keeping their security software up to date (1)<br>3. Policy and procedures effective in preventing an insider attack / a data breach (1)<br>4. Data protection laws in the resident country must be obeyed (1)<br>5. Backup and restore procedures (1)<br>6. Encryption (1) | | 2 |

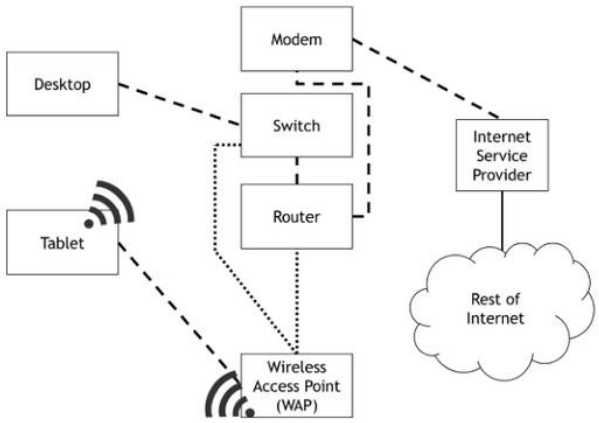
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | **Two** marks if completely correct.<br>**One** mark for any two correct cells.<br><br>**Robert** / **Zafer**<br>Washing machine design drawings: Read / Read<br>A folder of manuals for new machines: None / Read, Write Or Write, Read | Ignore spelling mistakes | 2 |

# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 2 of 5 · Question 2 part a · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c) | Award **one** mark from each category (maximum of 2)<br><br>Input:<br>• www.pearson.com / website domain (1)<br>• Uniform Resource Locator / URL (of Pearson website) (1)<br><br>Output:<br>• 2.20.38.113 (1)<br>• Pearson's machine IP(v4) address (1) | Don't penalise spelling or typos, especially in addresses | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d) | One mark for each dotted connection, correctly located. Maximum of 6 marks.<br><br>Note: WAP can be connected to either the Switch or the Router, but not both<br><br>  <br> | | 6 |

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 2 of 5 · Question 2 part a · 2 marks

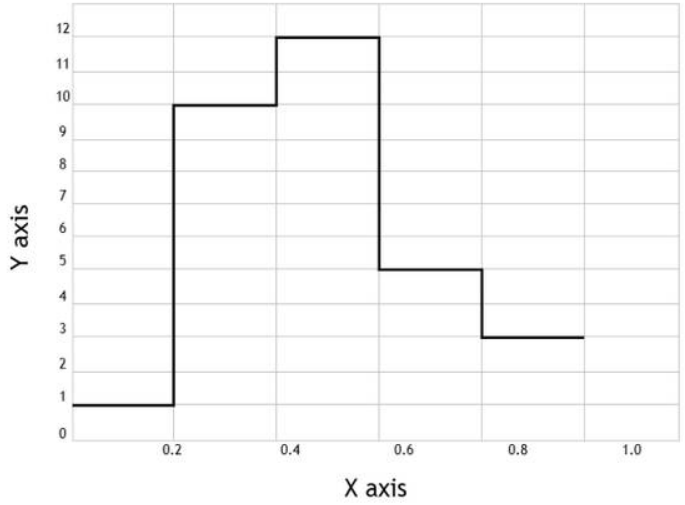
2019 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:<br><br>• At least four conversions plotted at the correct amplitude (1)<br>• Correct start point (1) 0 – 1 digital. 0 – 0 analogue<br>• Digital sound wave drawn (1) using candidate's plots<br><br><table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table><br> 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:<br><br>• Time (1)<br>• Sample interval/period (1)<br>• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:<br><br>• Amplitude / sound level / volume (1)<br>• Metres/centimetres/nanometres (1)<br>• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:<ul><li>Transfer time would be longer (1)</li><li>Larger file size / less compression (1)</li><li>Takes up more storage space (1)</li><li>Uses more of her data allocation (1)</li></ul></td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:<ul><li>Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)</li><li>Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)</li><li>Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)</li><li>Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</li></ul></td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:<ul><li>Storage host could be targeted by hackers (1)</li><li>Alyssa has less control over her files (1)</li><li>An untrustworthy employee (of the cloud storage provider) could steal her files (1)</li><li>Reliant on the storage provider for security / safekeeping (1)</li><li>Files could be intercepted/corrupted during upload/download (1)</li></ul></td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Cloud storage: what it is, pros, cons and security

Answer 3 of 5 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | ru2.mp3 | File/media/resource wanted (on the website)/ path / part of path (1) | | |
| | | | **Total for question 3** | **13** |

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 3 of 5 · Question 3 part b.ii · 2 marks

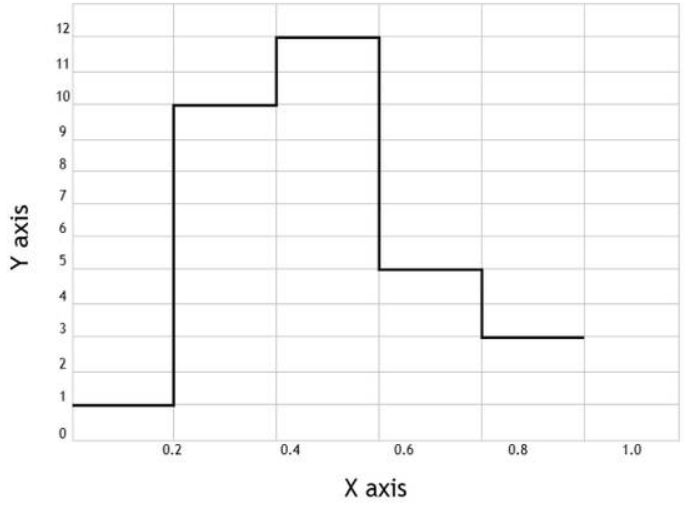
2022 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

| Question Number | Answer (flow chart replaced) | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Cloud storage: what it is, pros, cons and security

Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:<br><br>• At least four conversions plotted at the correct amplitude (1)<br>• Correct start point (1) 0 – 1 digital. 0 – 0 analogue<br>• Digital sound wave drawn (1) using candidate's plots<br><br><table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table><br> 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:<br><br>• Time (1)<br>• Sample interval/period (1)<br>• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:<br><br>• Amplitude / sound level / volume (1)<br>• Metres/centimetres/nanometres (1)<br>• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:<ul><li>Transfer time would be longer (1)</li><li>Larger file size / less compression (1)</li><li>Takes up more storage space (1)</li><li>Uses more of her data allocation (1)</li></ul></td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:<ul><li>Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)</li><li>Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)</li><li>Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)</li><li>Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</li></ul></td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
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Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | ru2.mp3 | File/media/resource wanted (on the website)/ path / part of path (1) | | |
| | | | **Total for question 3** | **13** |

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 4 of 5 · Question 3 part b.iii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

| Question Number | Answer (flow chart replaced) | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a) | Award **one** mark for each of:<br><br>Method<br>• Flowchart (1)<br><br>Justification<br>• It is a visual representation / does not use many words / does not rely on use of English language (1)<br>• It does not rely on understanding specific syntax (1)<br>• Overview without unnecessary detail (1)<br><br>OR<br><br>Method<br>• Pseudocode/structured English/written description (1)<br><br>Justification<br>• It does not rely on understanding specific syntax (1)<br>• It uses English language words (1) | Valid method must be specified first before mark for justification can be awarded. | 2 |

# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b) | Award **one** mark for any of the following:<br>• The test on line 3 has already limited the values to 1 and 2 (1)<br>• The only value left for inNum to be (on line 7) is 2 (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(i) | Award up to **two** marks for a linked description, such as:<br>• The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1) | Allow 4 to 0 instead of highest and lowest | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(ii) | Award **one** mark for any of the following:<br>• Reverse the (array of) scores (1)<br>• Copy the oldScores into newScores, in reverse order (1) | | 1 |

# Type MS · Cloud storage: what it is, pros, cons and security

## Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

| Question Number | Indicative Content | Additional Guidance | Mark |
|---|---|---|---|
| 6(d) | **Growth**<br>• Higher bandwidth available both on wired and wireless networks<br>• Reduced costs of large capacity storage devices<br>• Improved levels of encryption / authentication<br>• Increased trust in service providers<br>• Growth of service provider options<br>• Increased business models that aid scalability regarding storage capacity and costs<br><br>**Benefits**<br>• Sharing / collaboration on documents and setting of access permissions<br>• Access from any location connected to a WAN<br>• Reduced cost of local computing devices with less storage required<br>• Built in backups of data on server-based systems<br><br>**Drawbacks**<br>• Increased security risks incurred from outsourcing a service<br>• Relies on high-speed network connections<br>• Reliance on service providers continuing to provide a high level of service<br>• Increased chances of external data breaches<br>• Identifying responsibilities for data breaches / legal jurisdiction<br>• Increased power consumption from server systems and associated cooling systems<br>• Increased use of networks and network traffic which will have an impact on other network users<br><br>**Conclusion**<br>• A conclusion does not have to be stated explicitly, but can be inferred from the discussion | | 6 |

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 5 of 5 · Question 6 part d · 6 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

## Type MS · Cloud storage: what it is, pros, cons and security

### Answer 5 of 5 · Question 6 part d · 6 marks

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